

Experimental Protocol

Study: Hybrid Explainable Heterogeneous Graph Anomaly Detection for Healthcare Provider Fraud Detection

Dataset: CMS Healthcare Provider Fraud Detection Analysis (Kaggle / CMS LDS derivative)

Prediction unit: Provider (PotentialFraud $\in$ {No, Yes})

Document status: Methodology-ready for journal/conference submission

1. Research Hypotheses

All hypotheses are evaluated under **provider-disjoint stratified 5-fold cross-validation** on the training corpus. The Kaggle test split is reserved for optional post-hoc inference only (labels withheld).

ID	Hypothesis	Rationale	Primary test
H1	The proposed hybrid model (supervised R-GCN + Isolation Forest fusion) achieves statistically higher AUPRC than the best tabular baseline (CatBoost) and the best graph-feature baseline (RF + centrality).	Dual-signal design should capture relational structure and embedding-space anomalies missed by tabular models.	Paired comparison of fold-wise AUPRC (Hybrid vs. each baseline)
H2	Hybrid > GNN-only: fusing unsupervised anomaly scores improves recall and AUPRC over supervised R-GCN alone, especially for borderline fraud providers.	PotentialFraud is a proxy label; IF on embeddings adds robustness to label noise.	Ablation: Hybrid vs. R-GCN-only
H3	Hybrid > IF-only: supervised R-GCN improves precision over Isolation Forest alone on provider embeddings.	Pure anomaly detection ignores known fraud signals in training labels.	Ablation: Hybrid vs. IF-only
H4	A heterogeneous provider graph (provider–beneficiary–physician–diagnosis) yields higher AUPRC than a homogeneous provider-only graph with identical features.	Typed multi-entity relations carry fraud signal beyond provider aggregates (Yoo et al., 2023).	Ablation: Full heterograph vs. provider-only
H5	GNNExplainer subgraphs for flagged providers achieve explanation fidelity ≥ 0.70 and surface clinically interpretable motifs (shared patients, physician affiliation, diagnosis patterns).	Explainability must be graph-native and auditor-meaningful, not decorative.	Fidelity score + qualitative audit
H6	Provider collusion edges (shares_patient) incrementally	Medicare fraud often involves shared-patient	Ablation: w/ vs. w/o shares_patient

ID	Hypothesis	Rationale	Primary test
	improve recall without catastrophic precision loss.	billing rings (RDPGL, IJCAI 2023).	

Secondary (exploratory, not confirmatory):

H7: Fusion weight $\alpha \in (0,1)$ is strictly better than $\alpha \in \{0,1\}$ on validation AUPRC (confirms non-trivial hybrid benefit).

2. Dataset Split Methodology

2.1 Data sources (training corpus)

File	Role	Records (train)
Train-*.csv	Provider labels (PotentialFraud)	5,410 providers
Train_Beneficiarydata-*.csv	Beneficiary static features	138,556
Train_Inpatientdata-*.csv	Inpatient claims	40,474
Train_Outpatientdata-*.csv	Outpatient claims	517,737
Merged claims	Unified claim table	558,211

2.2 Label definition

- **Positive class (fraud):** PotentialFraud = "Yes" → 506 providers (9.35%)
- **Negative class (non-fraud):** PotentialFraud = "No" → 4,904 providers (90.65%)
- **Note:** Labels are provider-level, undocumented, and treated as **noisy proxy labels** (limitations acknowledged in §11).

2.3 Split policy

Split	Policy	Rationale
Primary evaluation	5-fold stratified CV on 5,410 train providers	Test labels withheld on Kaggle; CV is the only labeled evaluation
Per-fold internal split	Train 64% / Val 16% / Test 20% of providers within each outer fold	64+16=80% for outer-train; 20% held out as outer-test
Kaggle test set	Excluded from model selection and primary results	No PotentialFraud column in test labels file

Split	Policy	Rationale
Claim/beneficiary usage	All claims from providers in a fold's train set may construct graph edges; providers in val/test are masked at loss only	Graph includes cross-provider beneficiary bridges; provider labels never leak across outer-test

2.4 Preprocessing lockdown

All preprocessing statistics are **fit on outer-train providers only** per fold:

- Diagnosis/procedure vocabulary (top-K codes)
- Feature normalization (z-score, min-max)
- Beneficiary/physician frequency caps (UNK bucket)
- Isolation Forest (fit on **non-fraud** providers in outer-train only)
- Fusion weight α (tuned on **validation** only)

3. Provider-Disjoint Cross-Validation Protocol

3.1 Unit of splitting

Provider ID (Provider) is the sole unit for fold assignment. Claims, beneficiaries, and physicians appear in multiple folds only as **unlabeled graph context**; only provider nodes in the outer-test fold contribute to primary metrics.

3.2 Procedure

Input: $P = \{p_1, \dots, p_{5410}\}$ providers with labels y_p

Step 1: StratifiedKFold($n_splits=5$, $shuffle=True$, $random_state=42$)

→ preserve ~9.35% fraud rate per fold

Step 2: For each fold $k \in \{1, \dots, 5\}$:

Outer-test: T_k (~1,082 providers, ~101 fraud)

Outer-train: $R_k = P \setminus T_k$ (~4,328 providers)

Step 3: Split R_k into inner-train V_k (80%) and inner-val Val_k (20%)

(stratified on labels)

Step 4: Build graph G_k using claims from providers in $R_k \cup T_k$

(or R_k only for strict variant — see §11)

Step 5: Train on V_k ; tune α on Val_k ; report metrics on T_k

Step 6: Aggregate fold metrics: mean $\pm$ std

3.3 Leakage controls

Risk	Control
Provider label leakage	Never assign same provider to outer-train and outer-test
Feature leakage	Aggregates computed from outer-train claims only when used as provider input features
IF contamination	IF fit on non-fraud providers in inner-train only
Explanation leakage	GNNExplainer run post-hoc on frozen fold-k model; no test labels used in explanation selection
Random seed	Fixed seeds: 42 (CV), 42+k (fold k training)

3.4 Graph construction scope (default protocol)

Default (relaxed): Graph includes beneficiaries and physicians connected to outer-train **and** outer-test providers, but **loss and metrics are computed only on outer-test provider nodes**. This reflects transductive inference on a fixed claims universe.

Strict variant (appendix): Graph built from outer-train providers only; outer-test providers inserted at inference with inductive R-GCN. Report if results differ materially.

4. Baseline Models

Baselines mirror CMS fraud literature and directly test H1.

ID	Model	Input	Rationale (literature)
B1	Logistic Regression (LR)	Provider tabular aggregates	Linear baseline; FLAIRS/CMS studies
B2	Random Forest (RF)	Provider tabular aggregates	Strong classical baseline (MDPI Risks 2025)
B3	CatBoost	Provider tabular aggregates + categoricals	SOTA tabular on Medicare (Hancock & Khoshgoftaar 2021)
B4	RF + Graph Centrality	Tabular + degree, betweenness, closeness, eigenvector centrality	Primary graph baseline (Yoo et al., IEEE Access 2023)
B5	R-GCN (supervised only)	Heterogeneous graph	Isolates value of hybrid vs. GNN alone (H2)

ID	Model	Input	Rationale (literature)
B6	Isolation Forest (IF-only)	R-GCN embeddings (encoder frozen or untrained*)	Isolates value of hybrid vs. anomaly alone (H3)

*For B6 fair comparison: use **same trained R-GCN encoder** from B5 but **no classification head**; IF on embeddings. This ensures identical representation space.

4.1 Provider tabular features (B1–B4)

Aggregated per provider from outer-train claims:

- n_claims, n_patients, n_physicians, total_reimb, mean_reimb, std_reimb
- Inpatient claim ratio, mean claim duration, mean deductible
- Top-50 diagnosis/procedure code counts (inpatient + outpatient separately)
- Beneficiary demographics: mean age, gender ratio, mean chronic condition count

4.2 Centrality features (B4)

Computed on provider projection graph:

- Degree, betweenness, closeness, eigenvector centrality
- Optional: shared-patient count (without full hybrid pipeline)

5. Proposed Models

5.1 Primary model: Hybrid-RGCN-IF (full system)

Component	Specification
Encoder	2-layer R-GCN (HeteroConv + RGCNConv), hidden dim 64
Supervised head	Linear $\rightarrow$ sigmoid; weighted BCE (weight_pos = $N_{\text{neg}}/N_{\text{pos}}$)
Anomaly head	sklearn IsolationForest on provider embeddings
Fusion	$R_{\text{final}} = \alpha \cdot P_{\text{fraud}} + (1 - \alpha) \cdot \tilde{S}_{\text{anomaly}}$ where $\tilde{S}$ is min-max normalized IF decision function on val set; α tuned on inner-val
Threshold	Default 0.5; operating point selected on val to maximize F1 (reported separately)

5.2 Fallback encoder: Hybrid-SAGE-IF

If R-GCN exceeds 4GB VRAM or fails to converge within 1 week:

- Replace R-GCN with 2-layer GraphSAGE on homogeneous provider projection graph
- Edge attributes encode relation type counts

- All other protocol elements unchanged (report as appendix)

5.3 Heterogeneous graph schema (encoder input)

Node types: provider, beneficiary, physician, diagnosis

Edge

types: treated (ben→prv), affiliated_with (phy→prv), diagnosed (prv→dx), shares_patient (prv→prv, optional per H6)

Coarsening (4GB GPU): All 5,410 providers retained; beneficiaries and physicians capped at top-20,000 by claim frequency; remaining mapped to <UNK>; top-500 diagnosis codes.

6. Ablation Studies

Ablation ID	Variant	Tests	Expected outcome
A1	Hybrid (full)	H1 (primary)	Best overall AUPRC
A2	R-GCN only (B5)	H2	Lower recall vs. A1
A3	IF-only on embeddings (B6)	H3	Lower precision vs. A1
A4	Hybrid w/o shares_patient edges	H6	Recall drop if collusion matters
A5	Hybrid w/o diagnosis nodes	H4 partial	AUPRC drop if coding fraud signal lost
A6	Homogeneous provider graph + GraphSAGE	H4	Lower than heterogeneous R-GCN
A7	Fusion $\alpha=0$ (GNN-only)	H7	Worse than tuned α
A8	Fusion $\alpha=1$ (IF-only)	H7	Worse than tuned α
A9	Unweighted BCE vs. weighted BCE	Sensitivity	Weighted should improve recall

Reporting: Full ablation table in appendix; A1–A4 in main paper.

7. Hyperparameter Search Space

Search is **fold-inner validation only** (no outer-test peeking). Given resource constraints, use **grid search** (not exhaustive BO).

7.1 R-GCN encoder

Hyperparameter	Search space	Default
Hidden dimension d	{32, 64, 128}	64
Number of layers L	{2, 3}	2
Dropout	{0.1, 0.3, 0.5}	0.3
Learning rate	{1e-3, 5e-4, 1e-4}	1e-3
Weight decay (L2)	{1e-4, 5e-4, 0}	5e-4
Optimizer	Adam	Adam
Max epochs	{100, 200} + early stop	200, patience=15
Batch size (NeighborLoader)	{256, 512, 1024}	512
Neighbor sample sizes	{[10,10], [15,10], [25,10]}	[15, 10]
Loss	{weighted BCE, focal ($\gamma=2$)}	weighted BCE

Selection criterion on inner-val: maximize AUPRC.

7.2 Isolation Forest

Hyperparameter	Search space	Default
n_estimators	{100, 200}	200
contamination	{"auto", 0.0935}	0.0935
max_samples	{"auto", 256}	"auto"
Training subset	non-fraud inner-train providers only	—

7.3 Fusion

Hyperparameter	Search space	Default
α (GNN weight)	{0.1, 0.2, ..., 0.9} step 0.1	val-tuned
Normalization of IF score	min-max on val	min-max

7.4 GNNExplainer (explainability phase)

Hyperparameter	Search space	Default
Subgraph hops	{1, 2}	2
Explainer epochs	{100, 200}	200
Providers explained	Top-50 fraud + 10 FP + 10 TN	fixed set per full CV run

Compute budget: One hyperparameter configuration $\leq$ 30 min training on RTX 3050 4GB; total search $\leq$ 27 configs $\times$ 5 folds $\Rightarrow$ manageable with early stopping.

8. Evaluation Metrics

8.1 Primary metric

Area Under Precision-Recall Curve (AUPRC) on outer-test provider nodes per fold.

Justification: Severe class imbalance (9.35% fraud); AUPRC more informative than ROC-AUC (Bauder et al., 2019; Hancock et al., 2023).

8.2 Secondary metrics

Metric	Definition	Use
F1	Harmonic mean of precision, recall at threshold 0.5	Balanced reporting
Recall (Sensitivity)	$TP / (TP + FN)$	Fraud capture (investigator priority)
Precision	$TP / (TP + FP)$	False accusation cost
ROC-AUC	Threshold-agnostic ranking	Literature comparability
Recall@k	Recall in top-k scored providers ($k \in \{50, 100, 506\}$)	Operational audit budget
Precision@k	Precision in top-k	Complement to Recall@k

8.3 Efficiency metrics (appendix)

- Wall-clock training time per fold
- Peak GPU memory (GB)
- Inference latency per provider (ms)

8.4 Aggregation

Report **mean $\pm$ standard deviation** across 5 outer folds. Include **per-fold table** in appendix for transparency.

9. Statistical Significance Testing

9.1 Primary comparison

Hybrid (A1) vs. best baseline (highest mean AUPRC among B1–B4):

1. Obtain fold-wise AUPRC vectors: $a = [a_1, \dots, a_5]$, $b = [b_1, \dots, b_5]$
2. **Paired two-sided t-test** on fold differences $d_k = a_k - b_k$
3. Report: mean difference, 95% CI, t-statistic, **p-value**
4. Significance threshold: **$\alpha = 0.05$**

9.2 Multiple comparisons

When comparing Hybrid against **m = 4** baselines (B1–B4):

- Apply **Bonferroni correction**: reject if $p < 0.05/4 = \mathbf{0.0125}$
- Or report **Holm-Bonferroni** adjusted p-values (preferred)

9.3 Ablation comparisons

Paired t-test on fold-wise AUPRC:

- A1 vs. A2 (Hybrid vs. GNN-only) $\rightarrow$ H2
- A1 vs. A3 (Hybrid vs. IF-only) $\rightarrow$ H3
- A1 vs. A4 (w/ vs. w/o collusion edges) $\rightarrow$ H6

9.4 Effect size

Report **Cohen's d** on fold-wise AUPRC differences:

- $|d| < 0.2$ negligible; 0.2–0.5 small; 0.5–0.8 medium; > 0.8 large

9.5 Non-parametric fallback

If $n=5$ folds violate normality (Shapiro-Wilk on differences, $p < 0.05$):

- Use **Wilcoxon signed-rank test** (paired)
 - Report as sensitivity analysis in appendix
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10. Explainability Evaluation Methodology

Explainability is evaluated **post-hoc** on the **best Hybrid model** (highest mean AUPRC across folds). For each fold k , retrain on full outer-train R_k with best hyperparameters; explain selected providers from outer-test T_k .

10.1 Provider selection for explanation

Set	Size	Selection rule
True Positives (TP)	30	Highest R_final among correctly predicted fraud
False Positives (FP)	10	Highest R_final among non-fraud misclassified
True Negatives (TN)	10	Lowest R_final among correctly predicted non-fraud
Total	50	Fixed per fold → 250 explanations full CV

10.2 GNNExplainer protocol

1. Extract **2-hop ego-subgraph** around target provider (homogeneous projection for explainer compatibility)
2. Run GNNExplainer to obtain edge mask **M_E** and node feature mask **M_F**
3. Binarize mask at 0.5 threshold → subgraph **G'**
4. Record top-5 edges by mask weight

10.3 Quantitative fidelity metrics

Metric	Definition	Pass criterion (H5)
Explanation fidelity	$ f(G) - f(G') $	$/ f(G) $ where G' is masked subgraph
Sparsity	$ E' $	$/ E $ (fraction of edges retained)
Stability	Jaccard similarity of top-5 edges over 3 explainer seeds	Mean ≥ 0.60

10.4 Qualitative audit (motif taxonomy)

Two independent annotators (team members) code each explanation into motifs:

Motif code	Description
M1	High shared-beneficiary count
M2	Physician affiliation concentration
M3	Diagnosis code diversity / upcoding pattern
M4	Reimbursement outlier vs. subgraph neighbors
M5	No interpretable motif (explanation failure)

Report: motif frequency by TP/FP/TN; Cohen's κ for inter-annotator agreement.

10.5 Comparison to SHAP (appendix only)

Run TreeSHAP on CatBoost (B3) for same 50 providers. Compare whether SHAP cites **relational motifs** (M1–M3) or only tabular aggregates. Supports claim that GNNExplainer adds relational insight (Muhammad et al., 2025).

11. Threats to Validity

11.1 Internal validity

Threat	Severity	Mitigation
Label noise (PotentialFraud undocumented)	High	Frame IF as label-noise robustness; discuss LEIE limitation
Transductive leakage via shared beneficiaries across folds	Medium	Provider-disjoint splits; strict inductive variant in appendix
Hyperparameter overfitting (small n=5 folds)	Medium	Fixed search grid; report std; no test-set tuning
Fusion tuning on val	Low	Outer-test never used for α selection

11.2 External validity

Threat	Severity	Mitigation
Single-year CMS data (2009)	High	Acknowledge temporal/generalization limits
US Medicare only	High	Discuss NHIS/private insurance differences
Kaggle derivative, not raw CMS PUF	Medium	Cite dataset source; compare methodology to LEIE studies
No hospital entity	Medium	Scope claims to provider-centric fraud only

11.3 Construct validity

Threat	Severity	Mitigation
Provider label $\neq$ claim fraud	High	Explicit prediction unit statement; no claim-level claims
AUPRC on proxy labels	Medium	Supplement with Recall@k, case studies
Explanation fidelity $\neq$ ground truth	Medium	Qualitative motif audit; FP explanation analysis

11.4 Conclusion validity

Threat	Severity	Mitigation
Small fold count (n=5)	Medium	Paired tests + Wilcoxon fallback; report effect sizes
Multiple comparisons	Medium	Bonferroni/Holm correction
Hardware-constrained suboptimal HPO	Low	Report search space; release configs

12. Reproducibility Checklist

12.1 Code and data

- Public GitHub repository with MIT/Apache 2.0 license
- requirements.txt / environment.yml with pinned versions (Python 3.10+, PyTorch, PyG, scikit-learn, CatBoost)
- Fixed random seeds documented (42, 42+k)
- Dataset download script pointing to Kaggle rohitro/healthcare-provider-fraud-detection-analysis
- Preprocessing + graph construction scripts with CLI entry points
- Single-command reproduction: `python run_experiment.py --fold all --model hybrid`

12.2 Experimental artifacts

- Fold split indices saved as `splits/fold_{k}.json` (provider ID lists)
- Best hyperparameters per fold: `configs/fold_{k}_best.yaml`
- Raw fold-wise metrics: `results/metrics_fold_{k}.csv`
- Aggregated results table generator script
- Trained model checkpoints per fold (optional; ≥ 100 MB — use Git LFS or Zenodo)

12.3 Documentation

- README: hardware spec (RTX 3050 4GB), expected runtime ($\sim X$ hours full CV)
- Graph schema diagram + node/edge counts per fold
- Baseline reproduction instructions (CatBoost, RF+centrality)
- Explanation export format (GraphML/JSON subgraphs)

12.4 Paper reporting standards

- Report mean $\pm$ std for all metrics

- Report $n=5$ folds, fraud prevalence per fold
- Include confusion matrices at 0.5 threshold **and** val-optimized threshold
- Publish ablation table (A1–A4 minimum)
- Publish 3 full case-study subgraphs (1 TP, 1 FP, 1 TN)
- Limitations paragraph referencing §11 threats
- Code/data availability statement

12.5 Pre-registration (recommended)

- Upload frozen protocol to OSF/Zenodo **before** final fold evaluation
- Lock primary metric (AUPRC) and primary comparison (Hybrid vs. B4)

Methodology Section — Ready-to-Paste Template

4. Experimental Setup

We evaluate our framework on the CMS Healthcare Provider Fraud Detection Analysis dataset comprising 558,211 claims from 5,410 providers (506 fraudulent, 9.35%). Provider-level labels (PotentialFraud) serve as binary targets. Due to withheld test labels, all primary results use stratified 5-fold cross-validation with provider-disjoint outer splits (random seed 42). Each outer-train set is subdivided 80/20 for inner training and validation; fusion weight α and early stopping are tuned on inner validation only.

Baselines include Logistic Regression, Random Forest, CatBoost, and Random Forest with graph centrality features (Yoo et al., 2023). Our proposed **Hybrid-RGCN-IF** model encodes a heterogeneous graph (providers, beneficiaries, physicians, diagnoses) with a 2-layer R-GCN, trains a weighted BCE classifier, fits an Isolation Forest on non-fraud provider embeddings, and fuses scores via validation-tuned α .

Primary metric is AUPRC; secondary metrics include F1, recall, precision, ROC-AUC, and Recall@k. Statistical significance is assessed via paired t -tests across folds with Holm-Bonferroni correction. Explainability is evaluated with GNNExplainer on 50 providers per fold using fidelity, sparsity, and motif coding. Full hyperparameter search spaces, ablations, and reproducibility artifacts are documented in our supplementary protocol and public repository.

This protocol is calibrated for a **3-student, 1-month, RTX 3050 4GB** project while meeting expectations of applied ML and health informatics reviewers. If useful, I can next generate the **run_experiment.py CLI specification** or **LaTeX table templates** for Results §5.